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Computing

"ROOF COMPUTING With Standard Interfaces Harmonises The IoT ECOSYSTEM"

In order to ease the efforts of building applications over Roof platform, a unified interfacing and functional model is required. This is where Roof Computing, also known as IEEE P1931.1, comes in. Syam Madanapalli, Director - IoT Delivery, NTT DATA, talks about this technology in a conversation with Dilin Anand from EFY

Q. What is IEEE P1931.1?

A. IEEE P1931.1 is a federated networking and computational paradigm for the Internet of Things (IoT). It is always available for real-time on-site operation facilitation including next-hop connectivity for Things (IoT products), real-time context building and decision triggers. It provides efficient data connectivity to Cloud/service providers and always-'on' security for Things under Roof.

Q. How is it deployed?

A. Roof is implemented as a software platform on various devices that proxy IoT products and their IoT services to the rest of the world. The devices on which Roof is implemented include, but not limited to, mobile phones, home routers and other computing platforms. Roof is placed a few metres away from Things and far below Fog and Cloud.

Q. How does it help engineers working with the IoT?

A. Today, IoT ecosystem is vertically segmented, raising the entry for small players. Roof segments the IoT networks horizontally with standard interfaces between the physical and cyber-world. This enables the engineers and companies to play at their strengths and innovate. Device vendors, Cloud service providers and platform builders can focus on innovating with their products and services instead of working on end-to-end solutions.

Q. How does the technology work?

A. Roof is a platform for the edge and gateway devices and provides various application interfaces to interface with devices, cloud, service providers and users while preventing the intruders. The southbound interfaces allow the Things to be put under Roof in a secure manner and build trust with other Things under Roof. The northern interface helps in establishing a secure channel to Cloud and service providers to protect information and privacy, while feeding Cloud-based IoT applications. The horizontal interfaces help manage Things, and security and privacy. Security functions include authentication, authorisation and secure key establishment for exposing IoT services under Roof, while privacy function helps these manage access rights for data and IoT services.

Q. What do you feel about companies using IoT label for any Internet-connected device?

A. Most businesses depict the IoT and its value depending on the product or service they offer. The IoT deals with the physical world, and is deployed for a mission that requires it to function autonomously over a period of time, irrespective of what is happening elsewhere. For example, an IoT deployment in a factory should continue to function if there is a fibre cut outside the factory that disrupts Cloud connectivity. Things in the IoT should operate and cooperate in a secure and independent manner within the context of a local environment such as a home or factory. These need to connect to Fog or Cloud only for added value.

Q. What is Roof's biggest benefit to an engineer building IoT systems?

A. Ability to provide autonomous operations, secure communications and privacy by design are the fundamentals of Roof architecture. Roof with standard interfaces harmonises the IoT ecosystem and allows building more generic IoT gateways and Cloud services so that one can choose devices from any vendor and get the services from another vendor.

Q. How can it be used to securely manage devices?

A. Roof avoids exposing IoT devices directly over the Internet. It provides multi-level security, analyses threats and requires two-factor authorisation for suspected activities. It also supports authentication for multiple service providers securely to manage data access and privacy.

Q. How does it reduce the implementation barrier?

A. Roof can harmonise the IoT ecosystem by allowing individual components of the IoT system to evolve independently. This allows development of devices and other solutions, and taking these to the market.

In addition, Roof enables rapid development of innovative services based on existing devices by allowing access to multiple service providers. It provides the ability to choose and switch to any equivalent service provider. EFY